

CONTAINER: Bounding Credential-Compromise Impact in IEEE 2030.5 DER Systems

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Abstract

IEEE 2030.5 binds a distributed energy resource’s long-lived bootstrap identity to its operational authorization and forbids online certificate revocation. A stolen credential therefore retains control authority indefinitely, and no online mechanism can withdraw it. We present CONTAINER, which replaces the conflated device credential with attested, short-lived, scope-bound operational credentials and uses non-renewal rather than revocation as the containment mechanism, deployable incrementally through native and gateway-proxy tiers. Because reachability alone does not distinguish these designs, we measure impact along four dimensions — cyber reach, state-dependent commandable flexibility, retained authority, and feeder consequence — and give a worked example of two compromises with identical reach but divergent flexibility, persistence, and feeder outcome. We implement a credential service with real X.509 and mutual TLS and a real OpenDSS feeder pipeline. On the IEEE 8500-node feeder, ten implemented security invariants pass and attested issuance costs about 0.23 ms. Our central physical finding is negative and constrains the design space: sweeping the authorized envelope continuously, *no* reactive scope both contains a compromise and delivers the volt-var service IEEE 1547-2018 requires. At the full Category B envelope the induced overvoltage area is 87.9 p.u.-node with every seed tripping the overvoltage screen, and even at zero au-

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thorized reactive injection it is 31.2 with every seed tripping, because suppressing the absorption the feeder relies on is itself an attack. Scope having failed, lifetime is the remaining control: across twenty paired seeds a compromise at the conformant envelope holds the feeder in violation for the whole horizon under a long-lived credential, whereas under CONTAINER the feeder returns to its pre-attack baseline within one minute of expiry with a settled residual of zero. We show that integrated harm is the product of a measured accrual rate and the credential lifetime, so the containment factor is a lifetime the operator chooses rather than an effect size we measured, and we report its cost: a 51.5% rise in regulator tap operations. These are reduced-scale pilot results, reported as exploratory; the pre-registered confirmatory sweep is future work.

Keywords: Critical infrastructure protection, distributed energy resources, IEEE 2030.5, remote attestation, authorization, cyber-physical security

1. Introduction

Distributed energy resources now participate in grid operations at a scale that makes their control interface a systemic concern. IEEE 2030.5 (Smart Energy Profile 2.0), profiled for distribution use by the Common Smart Inverter Profile (CSIP), is the application protocol through which utilities and aggregators dispatch inverter-based resources. An adversary who obtains a credentialed identity in this ecosystem does not merely read data; it can command real and reactive power on physical hardware connected to the grid.

Two properties of the IEEE 2030.5 trust model make this worse than a conventional credential-theft problem. First, the device’s bootstrap identity and its operational authorization are the same long-lived certificate: there is no separate, short-lived operational credential to lose. Second, the standard forbids certificate revocation; it directs that certificate authorities not maintain revocation lists or run status responders, and that clients and servers not check them [1]. A stolen credential is therefore valid until it expires, and its lifetime may be indefinite.

A natural objection is that this is a solved problem: model the DER authorization surface as a privilege graph and measure blast radius as reachability, as a mature line of work does for enterprise networks [2, 3, 4]. We argue this reduction is insufficient as ordinarily applied. What determines the damage of a DER compromise is not only which nodes an adversary can reach, but how much physical capacity those nodes command, how long the adversary retains that capacity, and what the feeder does in response under the grid state at the time. Reachability over a static privilege graph is a function of the attack locus and the graph alone; the quantities that matter are functions of the defensive policy and of exogenous physical state, and neither is topological.

Contributions

- *CONTAINERDER*, an incrementally deployable credential redesign for IEEE 2030.5 using attested, short-lived, scope-bound operational credentials, with non-renewal as containment and explicit session- and command-effect enforcement, deployable through native and gateway-proxy tiers.
- A *working prototype* — a credential service with real X.509 issuance, mutual TLS, and renewal, plus an OpenDSS feeder pipeline — with ten implemented security invariants verified and containment timing measured.
- A *negative result that constrains the design space*: on the IEEE 8500-node feeder [5], an authorization scope narrow enough to contain a compromise is too narrow to deliver the volt-var service IEEE 1547-2018 mandates and CSIP programs procure. An adversary confined to the *conformant* volt-var envelope still drives the feeder past an IEEE 1547 overvoltage trip threshold. Scope narrowing alone therefore does not contain, and the apparent adequacy of a narrow access list in our earlier configuration was an artifact of an envelope too small to deliver the service at all.
- A *pilot evaluation* with paired per-seed contrasts over twenty seeds showing that, at that conformant scope, bounding the *duration* of retained authority is the only remaining control, and characterising it honestly: the

mechanism truncates the exposure window rather than attenuating the excursion, so integrated harm is a measured accrual rate multiplied by the lifetime and the containment factor is a lifetime the operator chooses, not an effect size we measured. We report what it costs — a 51.5% rise in regulator tap operations — and where it does not pay, and we quantify the containment latency it buys: expected retention is the credential lifetime divided by the per-cycle attestation detection probability, not the lifetime itself.

The threat model separates bootstrap identity, operational credential, and authorization scope, and names the standard’s conflation of the first two as the driver of unbounded persistence.

We deliberately do *not* claim the four-dimension accounting of Section 3 as a contribution. Adding resource-by-scope product nodes, a time index, and attached state and policy to a classical attack graph yields it, and the separation propositions it supports are true by construction, since a reachability-only measure cannot depend on variables that are not among its arguments. We keep the notation because the paper needs a vocabulary for how much, for how long, and under which grid conditions. It is not a theoretical advance, and the load-bearing claims here are empirical.

We state scope plainly. The engine, credential service, and feeder pipeline are implemented and covered by a regression suite. The evaluation is a pilot: two public feeders, one solver, one workstation, and — at the conformant scope envelope where the central result lies — one penetration level and two operating states. The pre-registered confirmatory sweep, with a cross-solver check, constrained hardware, a stealth-constrained attacker, and the full seed budget, is future work.

2. Background and Threat Model

2.1. Authorization in IEEE 2030.5 / CSIP

IEEE 2030.5 uses mutual TLS with X.509 device certificates. A device's identity is the certificate-derived LFDI/SFDI, and its authorization is scoped by a `FunctionSetAssignments` resource that binds it to one or more `DERPrograms`. Physical control authority is carried by the `DERControlBase` control modes: active-power capping (`opModMaxLimW`), signed fixed active power for storage charge and discharge (`opModFixedW`), reactive power (`opModFixedVar`, `opModVoltVar`), volt-watt and frequency-watt curves, and the hard connect and energize controls [6]. The standard specifies certificate-based authentication and function-set scoping; it does not mandate a single canonical access-control object.

2.2. Certificate validity and revocation

The trust model has two properties central to this work [1]. Device certificates have indefinite lifetime, and revocation is prohibited: certificate authorities are directed not to maintain revocation lists or run status responders, and endpoints are directed not to check them. The stated reasons against online revocation are concrete design constraints, not incidental: field devices have no guaranteed connectivity, a hard-fail check destroys availability while a soft-fail check is trivially bypassed by blocking the responder, revocation presupposes a secure time source that inverters may lack, and no party is defined as responsible for reporting a compromised device. The sanctioned substitute is a local allow or deny list. The consequence for our threat model is direct: once a credential is compromised, there is no online mechanism to withdraw it, so its authority persists until it expires.

2.3. Three separated layers

We separate three layers that the baseline standard conflates. *L1, bootstrap identity*: the hardware-rooted device certificate. *L2, operational credential*: the

credential that authorizes live sessions to an aggregator or DER management system. *L3, authorization scope*: the server-side binding from an identity to the function sets it may invoke. In baseline IEEE 2030.5, L1 and L2 are the same long-lived certificate, with no attestation gate and no revocation path. This conflation, not any single cryptographic weakness, is what makes a compromise persist for the certificate lifetime.

2.4. *Compromise loci and adversary*

We consider four compromise loci of increasing consequence: a single device, an edge gateway, an aggregator or DER management system, and a utility-side issuing role. The adversary obtains a valid credential at one locus (by key extraction, session theft, or a misconfigured or compromised server granting scope) and issues protocol-conformant control commands within that credential’s authorization. The adversary does not break TLS or the underlying cryptography. The utility-side issuing role is the ceiling on what any credential discipline can bound, because it can subvert issuance itself; we treat it as the boundary of the design’s protection rather than a case the design defeats.

2.5. *Credential lifecycle and existing containment*

The motivation depends on exact certificate-management behavior, which differs across editions: IEEE 2030.5-2018 [7], IEEE 2030.5-2023 [8] (which supersedes it, published December 2024), and the profile in force [9]. We separate four questions the literature tends to conflate: whether a certificate can be cryptographically revoked; whether a server can locally deny an identity despite a valid certificate; whether a device can be re-enrolled or re-keyed; and whether an active session or an already-issued control remains effective after credential expiry. Lack of CRL/OCSP bounds only the first.

The paper’s primary open risk, stated plainly.. **We were unable to obtain IEEE 2030.5-2023.** It is paywalled at roughly US\$255, is not in the IEEE GET free-standards programme, and we found no publicly accessible document quoting its clause text — every free source that quotes 2030.5 quotes the 2018

Table 1: Every standards claim the argument uses, the accessible source of record for it, and what remains unverified. **No row has been checked against IEEE 2030.5-2023**; the edition in force could not be obtained (see text). SAND = Sandia SAND2019-1490 [1]; CSIP = Common Smart Inverter Profile Implementation Guide v2.1 [9]. “Free source” means a document a reader can check without purchase.

ID	Claim	Free source of record	Residual risk
S1	Device certificate lifetime indefinite	SAND §4.1, verbatim	2023 clause unread
S2	CRL use prohibited	SAND §4.1, verbatim	2023 clause unread
S3	OCSP prohibited	SAND §4.1, verbatim	2023 clause unread
S4	Local allow/deny listing available	SAND §4.1, §3.1.1.3; CSIP §5.2.1.4	permitted, not required (SAND §3.1.1.4)
S5	Certificate replacement / re-enrollment possible	none	unsupported ; SAND Rec. 5 implies absent
S6	LFDI/SFDI derives from certificate	SAND §3.1.1.1 and CSIP §5.2.1.2 (independent)	low ; two sources agree
S7	FunctionSetAssignments scopes operations	CSIP §5.2.3.1	clause number only
S8	Scheduled DER controls can persist	CSIP §4.4.2, §6.1.8.2	high ; 2023 reportedly drops mandatory DER-Control modes

edition. Our premise, that a certificate’s lifetime is effectively indefinite while online revocation is unavailable, is therefore evidenced against the *2018* edition and the Sandia analysis [1], not the edition in force. This is the single largest threat to the paper’s external validity and we do not minimise it. Table 1 gives each claim’s evidence row by row. Two rows matter most. **S5 is unsupported by any source we could access** and contradicted in spirit — Sandia lists dynamic re-provisioning as a recommendation, i.e. a capability judged absent — so we treat re-enrollment as something CONTAINERD must supply. **S8 is most exposed to edition drift**, since the 2023 revision is described as eliminating mandatory DERControl modes, exactly S8’s territory.

2.6. Assets, adversary, and goals

Protected assets: DER control integrity, fleet authorization, operational and attestation keys, and feeder stability and service continuity. *Adversary capabilities:* steals an operational key (and, where tested, a bootstrap key), compromises firmware, steals an active session, compromises a gateway or aggregator, issues protocol-conformant commands, and observes authorized telemetry. *Exclusions:* breaking standard cryptography, physical destruction, and compromising both the root issuer and all audit controls, which we treat as the protection boundary rather than a defended case. *Security goals:* least authority, bounded new-session authority, bounded active-session authority, bounded command-effect duration, fresh-key containment, replay resistance, safe degradation, and auditability. *Non-goals:* defending against malicious root issuance without added governance, detecting every hardware attack, replacing grid protection, and proving safety for every feeder.

3. Impact Accounting, and Why Reachability Alone Cannot Rank These Designs

This section fixes notation and says why we need it. It is *not* a contribution. Adding resource-by-scope product nodes, a time index, and attached state and policy to a classical attack graph yields exactly the accounting below, so an extended attack graph [10, 3, 11] agrees with us rather than competing. We keep the notation only because the experiments in Section 8 move “how much”, “for how long”, and “under which grid conditions” differently, and one scalar cannot express that.

Substrate and the four quantities.. A static privilege graph $G = (V, E)$ has principals and resources as nodes and access relations as edges; $x \subseteq V$ is a set of compromised credentials, and a policy $\pi = (L, \text{Att}, A)$ gives a lifetime function L , an attestation-gated renewal predicate Att , and a scope map A . The device and grid state u_t and the feeder power-flow and protection model F appear nowhere in G , which is the point.

Cyber reach $\text{BR}_{\text{reach}} = \text{ReachSet}(x, \pi)$, a count, is the least fixed point of the π -enabled access relation from x : the endpoints the compromised authority can send an authenticated request to. *Commandable flexibility* BR_{flex} , in kW and kVAr, is the aggregate over reachable DERs of the feasible set-point region $\mathcal{C}_d(t; u_t, A_\pi(d))$ given irradiance, state of charge, and inverter limits; a read-only scope yields the current set-point alone, hence zero. *Retained authority* separates three durations, in seconds,

$$\text{BR}_{\text{auth}} = \max(T_{\text{cred}}, T_{\text{sess}}, T_{\text{cmd}}), \quad (1)$$

where T_{cred} runs until the stolen credential is refused for a *new* session, T_{sess} until already-authenticated sessions can no longer issue commands, and T_{cmd} until queued or latched controls stop affecting the DER; the capacity-time exposure $\text{BR}_{\text{exp}} = \int \text{BR}_{\text{flex}}(t) \mathbf{1}[\text{authority effective}] dt$ carries kW·h or kVAr·h. This split is the design's load-bearing one: expiry bounds T_{cred} only. *Physical consequence* $\text{BR}_{\text{phys}} = \max_a J_V(F, u, a)$ is the worst outcome reachable within the adversary's authority, where J_V is the voltage-violation area outside the ANSI C84.1 band weighted by magnitude. BR_{phys} carries two units and we label every table with which: *p.u.-node* evaluated at one operating point, and *p.u.-node-min* accumulated over a horizon. Secondary metrics are maximum per-unit deviation, violating-node count, protection trips, regulator tap operations, energy not served, unnecessary curtailment, and recovery time.

Three separations, true by construction.. Call a measurement *reachability-only* if it is a function M of $\text{ReachSet}(x)$ alone. Then: varying only the scope map A at fixed reachable set changes BR_{flex} ; an attestation-gated ephemeral credential and a long-lived unrevoked one differ in BR_{auth} and BR_{exp} at equal initial reach; and at fixed x and π the same authorized command is harmless at one operating state and drives nodes past an IEEE 1547 overvoltage trip threshold at another. **None of these can fail.** M is single-valued in $\text{ReachSet}(x)$ and each construction varies a quantity that is by definition not an argument of M , so they restate a definition rather than establish a result. We state them

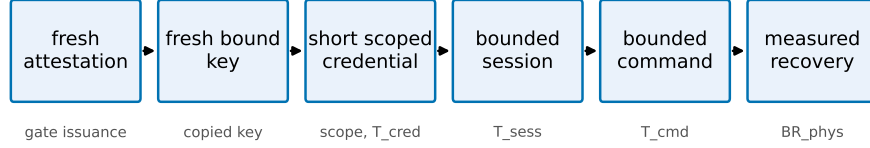


Figure 1: The CONTAINER containment chain. Each link bounds a dimension: attestation gates issuance, fresh-key binding contains a copied key, scope and short lifetime bound BR_{flex} and T_{cred} , session and command enforcement bound T_{sess} and T_{cmd} , and measured recovery bounds BR_{phys} .

because they say which quantities an evaluation must move, and each ships as a regression test in the released engine. **They are not evidence for the design; that is in Section 8, and only there.** One correction: the third construction previously turned on a reverse-power protection trip the simulator could not represent, there being no protection model at all. Section 6 now implements the IEEE 1547-2018 Category II overvoltage screen, and Section 8 reports the state dependence as measured.

4. CONTAINER Design

CONTAINER separates the layers that IEEE 2030.5 conflates and enforces credential lifetime at the session and command layers, not only at issuance. The design is a containment chain: fresh attestation, a fresh hardware-bound key, a short scoped operational credential, a bounded session, and a bounded command effect, so that removing authority actually removes control. Each link targets a dimension of the model (Fig. 1).

4.1. Components and trust layering

The device keeps a long-lived, hardware-rooted bootstrap identity [12] used only for enrollment and attestation, never accepted directly for DER control. An *attester* produces fresh evidence bound to a freshly generated operational public key; a *verifier* checks evidence against endorsement and reference values

and returns an explicit result; a *credential issuer* signs a short-lived operational certificate embedding role, site, scope, and expiry only for a valid attestation result; an *authorization service* maps the operational identity to least-privilege function-set assignments; the *2030.5 server or proxy* authenticates the operational certificate, enforces a maximum session age, and rechecks authorization per command or at a bounded interval; a *session controller* and a *command-lifecycle controller* bound authority beyond expiry; and a *risk controller* selects deny, scope narrowing, or bounded emergency renewal. We use the Entity Attestation Token [13] under the RATS Attester/Verifier/Relying-Party model [14], and define a concrete EAT profile rather than treating EAT as a complete attestation protocol.

4.2. Fresh-key binding

Attesting a device is not sufficient: a healthy device can pass attestation while an attacker retains a previously copied operational key. CONTAINERD therefore requires that every credential epoch use a newly generated operational key, generated in a TPM, TEE, or secure element where available and marked non-exportable, whose public-key digest is carried in the attestation evidence. The issuer signs only the attested public key, refuses renewal that presents only a prior certificate, rejects replayed evidence and reused nonces, and rotates the key even when the device stays healthy. This is what contains a copied key across the next epoch.

4.3. Session and command-effect enforcement

Certificate expiry does not end an open TLS session or an already-issued long-duration control. CONTAINERD closes sessions at credential expiry (or bounds session age to the credential TTL) and revalidates authorization per command, and it bounds adversary-issued controls through a maximum command duration, cancellation on containment, and safe-default restoration, with explicit handling of scheduled DERControls. The two quantities the prototype

Table 2: Safe-mode authorization states.

State	Allowed	Denied
Healthy	normal scoped control	out-of-scope control
Verification delayed	read telemetry, hold last safe schedule	new high-impact dispatch
Attestation uncertain	read-only, bounded emergency control	arbitrary set-points
Known compromise	read-only or total denial	all control
Issuer outage	prior low-risk scope, bounded grace	scope expansion, high impact

measures are

$$\Delta_{\text{expiry}} = t_{\text{last adv. command}} - t_{\text{expiry}}, \quad (2)$$

$$\Delta_{\text{effect}} = t_{\text{effect ends}} - t_{\text{expiry}}, \quad (3)$$

each targeted to zero or a small declared bound. These are exactly the T_{sess} and T_{cmd} components of BR_{auth} .

4.4. Safe-mode policy

Failure behavior is explicit rather than fail-open. The risk controller chooses among the states in Table 2; no failure mode grants indefinite authority, and the safety of each degraded state is validated under feeder conditions rather than assumed.

4.5. Migration compatibility

We separate three compatibility levels, and the protocol requires a passing interoperability test before any of them is claimed, rather than asserting “backward compatible.” None has yet been tested; Section 8 reports interoperability as untested. *Protocol compatibility*: 2030.5 messages remain standard conformant. *Implementation compatibility*: unmodified client or server software

can participate. *Deployment compatibility*: legacy and upgraded devices coexist. Three tiers realize this: a bootstrap-only legacy tier, a gateway-attested proxy tier where an edge gateway holds the operational credential for downstream legacy devices, and a native device-attested tier. We claim only the compatibility level demonstrated by an interoperability test; if only the proxy tier interoperates with an unmodified endpoint, we say CONTAINER supports incremental deployment through a credential-terminating gateway proxy, and no more.

4.6. What non-renewal inherits from the constraints it cites

Section 2 adopts the Sandia analysis’s reasons why online revocation fails for DER: no guaranteed connectivity, no trustworthy time on inverters, and no responsible reporting party. Our mechanism inherits two of the three, and we state that rather than let a reader find it.

Secure time. Expiry is a judgment about time, so an endpoint that cannot trust its clock cannot enforce a short lifetime — the deficiency cited against OCSP. An adversary who skews a device clock would extend a “short-lived” credential indefinitely. This is a potential break of the core mechanism, so **expiry is never enforced by the endpoint alone**: the issuer refuses renewal server-side where a trustworthy clock exists; each renewal binds a fresh key to a verifier-supplied nonce, so replaying an attestation cannot extend authority however the clock reads; and session and command limits are monotonic counters, not wall-clock deadlines. A skewed clock then costs availability, not containment. We have not measured skew.

Connectivity. A twelve-hour lifetime means a DER losing backhaul for twelve hours loses control authority, and rural outages of that length are routine. Our availability figures come from in-process fault injection, not networked outages, and renewal storms are untested. An aggregator dropping a fifth of fleet controllability during a verifier outage has created a reliability hazard while closing a security one. Safe-mode degradation to an autonomous volt-var profile exists for this reason; whether that trade is acceptable is an operator’s call.

Who reports compromise. Attestation detects only what it measures. Firmware integrity is measurable; **a stolen operational key on a device with intact firmware passes attestation**. Per-epoch fresh-key binding answers this, but containment latency is then not the lifetime: with per-cycle detection probability p and lifetime T , expected retention is T/p — twelve hours at $T=6\text{ h}$, $p=0.5$, and equal to T only at $p=1$. **We report T/p as the containment-latency figure of merit**. The alternative, a narrow access list on a long-lived credential, contains a copied key only when someone notices and edits the list, which has no bounded latency at all. Bounded beats unbounded; it does not beat instant.

Which tier is deployable. Presenting three tiers evenhandedly is misleading: only one is deployable within the next standards cycle. The native tier needs attestation and a trusted clock on constrained inverter hardware; the legacy tier changes nothing. **The gateway-attested proxy tier resolves all three inheritances at once** — the gateway holds the credential and the clock, attestation runs on capable hardware, downstream devices stay unmodified, and the gateway buffers short outages — and it alone requires no change to already-certified endpoints. We regard it as the deployment path.

4.7. Implementation status

The design is realized as an issuance and renewal service using real X.509 certificates and mutual TLS, with a Python IEEE 2030.5-style client in place of the EPRI C client and structured event logging for every credential, session, and command transition. Its security invariants and workstation overhead are reported in Section 8; a native EPRI-client integration and constrained-hardware measurements remain future work.

5. Cyber-Side Analysis Engine

We implement the cyber-side impact dimensions as an analysis engine, `pkimodel`, that computes BR_{reach} , BR_{flex} , and BR_{auth} for a compromise locus under a policy. The engine models the identity and delegation graph over

the six node types of the threat model, with typed issuance, delegation, and authorization edges. Its central modeling choice follows the formalism: a credential’s reach (which DERs it can address) is represented independently of its scope (which function sets it may invoke at each DER). This is what makes the separating counterexample expressible in code, and we ship that counterexample as a regression test: the two compromises resolve to an identical reachable set of eleven DERs, yet the engine computes zero commandable kW for the attested credential against tens of MW for the legacy one, and a persistence of hours against the analysis horizon.

The authorization policy is parameterized, not fixed: a hardcoded permissive access model would overstate what a stolen certificate can command, so the engine exposes four ACL realism levels spanning single-device, site, feeder-segment, and whole-fleet scope, varied in the evaluation. On a synthetic 200-device fleet over four feeders the reachable-DER count grows from one (single-device and site) to fifty (feeder) to two hundred (fleet), with commandable capacity in step; the feeder-level figure is the fleet divided by the feeder count and moves with that parameter. This operationalizes RQ1: most reach gained by a compromise is attributable to authorization scope rather than to authentication alone. Retained authority is returned as a distribution over duration, not a point estimate, and separates credential, session, and command-effect components.

The engine consumes synthetic scenarios from a deterministic, seedable generator, since real utility authorization graphs are not public; the generator is released as part of the artifact. The engine is cyber-side only. It produces the reachable-DER set and the commandable-capacity envelope that the feeder co-simulation consumes to compute BR_{phys} .

6. Feeder Simulation Methodology

The physical consequence dimension is measured by a feeder simulation that translates an authorized-control reach from the analysis engine into a malicious

dispatch program and runs it on a public distribution feeder. We take the IEEE 8500-node feeder [5] as the primary benchmark and the PNNL 9500-node system [15] as an advanced validation tier. Time-series inputs come from public solar and load datasets. This part of the work is methodologically routine relative to the false-data-injection literature on distribution systems, so we build it to be unimpeachable rather than novel.

This is a single-solver study, and we state the consequence rather than the aspiration: the authorization envelope is applied directly in the OpenDSS driver, so the constraint a credential expresses is enforced on the dispatch that actually reaches the feeder, but no second solver corroborates the result. Section 8 reports what that costs. The attack families are volt-var curve distortion, volt-watt curve distortion, export-limit distortion, and curtailment spam. The metrics are the count of nodes violating the ANSI C84.1 bands [16], maximum voltage deviation, the area under the violation curve over time, regulator tap operations, unnecessary curtailment, energy not served, and recovery time to steady limits, with the per-unit voltage excursion as the primary continuous measure and a protection-trip flag as the discrete outcome.

Four modelling choices, three of them corrections.. DER model. The fleet is OpenDSS `PVSystem` objects with an inverter rating and an `InvControl` volt-var controller on the IEEE 1547-2018 Category B default characteristic [17]. An earlier version used `Generator` objects at fixed P, Q , which have no inverter and no curve, so a “volt-var curve distortion” attack was really a static setpoint delta and the attack-family names overstated what was simulated. Curve distortion is now literal: the adversary rewrites the characteristic.

Counterfactual. The base case dispatches PV at its *legitimate scheduled output* under that curve. An earlier version set every unit to $P=Q=0$, so the counterfactual feeder had no DER at all and ΔJ_V conflated “connect this much generation” with “a credential was stolen”. All ΔJ_V here are against legitimate operation, and we report the legitimate base case’s own violation area so penetration-driven stress is visible separately.

Control mode. We solve with `controlmode=static`, which does *not* disable the feeder’s controls, and we release the diagnostic that shows it (`results/controlmode.json`). Reaching the legitimate base case at light load moves 8 of the 12 regulators, for 32 tap steps, and changes 6 capacitor states; a full-scope attack then moves 11 regulators for a further 24 tap steps and 3 capacitor changes. Controls are inert only under `controlmode=off`, where the same attack drives the total overvoltage area to 998.0 p.u.-node and $v_{\max}$ to 1.3145 p.u., against 133.1 p.u.-node and 1.1282 p.u. with controls active — the feeder’s own regulation absorbs a factor of 7.5. Both figures in that comparison are total areas, not attack-minus-base differences, so that they are commensurable. Regulators and capacitors therefore act against the disturbance in every result reported, and we report tap operations throughout, including where CONTAINER increases them.

Protection. Every node is screened against the IEEE 1547-2018 Category II overvoltage trip thresholds (OV2 1.20 p.u./0.16 s; OV1 1.10 p.u./2.0 s), plus a substation reverse-power flag. Earlier versions declared a trip metric with no protection model behind it, so it could never fire and any claim resting on it was unsupported.

We do not tune feeder parameters to amplify impact; we report what the standard models produce. The clean base case, the four attack families, the eight metrics, and the cross-solver agreement are the acceptance criteria for this component. As with the credential service, a first benchmark run on the IEEE 8500-node feeder appears in the pilot evaluation; the full sweep on both feeders, with the GridLAB-D cross-check and seasonal operating states, follows the pre-registered protocol chosen in advance.

7. Evaluation Protocol (Pre-Registered)

No measured results are reported in this section. We state the evaluation protocol for the confirmatory sweep, and we state precisely what our immutability evidence for it is worth, because a pre-registration that overstates its own guar-

antee is worse than none.

The scenario matrix and statistical plan are recorded in `PREREGISTRATION.md` with a SHA-256 content hash recomputable by `validate_matrix.py`. **That hash is not a timestamp.** It was computed by the same party who could have edited the file, so it shows only that the matrix has not changed since the hash was written; it cannot show *when* the matrix was fixed relative to seeing any outcome. The repository’s version-control anchor does not repair this: the initial commit capturing the pre-registration is dated 2026-07-30, which **postdates every pilot result reported in Section 8**. We therefore make no claim that the design could not have been adjusted after seeing pilot outcomes. It could have been, and no artifact in this repository proves otherwise.

The consequence follows directly, and we adopt it without qualification: the measured results in Section 8 are reported as **exploratory**, not confirmatory, and are excluded from the confirmatory tests.

Baselines and mechanism ablations.. Six credential-lifecycle designs are compared: legacy identity-as-authority (B1), legacy with a narrow access list (B2), ephemeral-only (B3), attestation-only (B4), full CONTAINER (B5), and safe-mode renewal (B6). Each baseline is specified by explicit fresh-key, session-enforcement, and command-cleanup attributes, because the central claim is that short lifetime alone does not contain a copied key, an active session, or a persistent command. Eight mechanism ablations (A1–A8) each remove one link of the containment chain, so the measured reduction is attributed to a mechanism rather than to shorter certificates. Table 3 gives the baseline mechanism attributes.

Factors and design.. The factors are credential time to live, authorization scope, compromise locus (with the issuer treated as a trust boundary, not a defended case), DER penetration and technology mix, attack program, attestation-trust state, renewal policy, and operating state. Every attack runs under both a worst-case optimizing attacker and an operationally realistic telemetry-limited

Table 3: Credential-lifecycle baselines. FK = fresh key per epoch; Sess = session enforcement; Cmd = command-effect cleanup.

ID	Design	Lifetime	Attestation	Scope	FK	Sess	Cmd
B1	legacy identity-as-authority	long	no	deployment ACL	no	no	no
B2	legacy + narrow ACL	long	no	narrow	no	no	no
B3	ephemeral-only	short	no	reference	opt.	expiry	opt.
B4	attestation-only	long	yes	reference	enroll	weak	no
B5	full CONTAINER	short	yes	narrow	per-epoch	yes	yes
B6	CONTAINER safe-mode	short	yes	narrow/degraded	per-epoch	yes	yes

attacker, so no result is only an omniscient upper bound. The primary comparison is the fully-crossed lifetime-by-scope factorial, whose interaction term is the compositional claim.

Statistics.. The experimental unit is the complete paired scenario run, not the per-timestep sample. The primary voltage-violation area is analyzed with a mixed-effects model, $\log(1 + Y) = \beta_0 + \beta_1 \log(\text{TTL}) + \beta_2 \text{Scope} + \beta_3 \log(\text{TTL}) \times \text{Scope} + b_{\text{day}} + b_{\text{seed}} + \epsilon$, whose interaction coefficient tests the primary hypothesis, with robust, aligned-rank, or permutation fallbacks where assumptions fail. Protection trips use mixed-effects logistic regression — now a well-defined outcome, since Section 6 implements a protection model — violating-node counts a negative-binomial mixed model, and recovery time survival analysis. Effect sizes are the median paired difference, ratio of geometric means, paired rank-biserial correlation, and odds ratios, each with bootstrap confidence intervals, with Holm correction within rather than across hypothesis families. A five-seed pilot and a power analysis set the production seed count before final comparisons are viewed. These procedures apply to the confirmatory sweep. The pilot in Section 8 reports paired per-seed contrasts with bootstrap intervals, which its stored data do support, and does not report the mixed-effects estimates, which they do not.

8. Pilot Evaluation

We report a pilot evaluation that exercises the full pipeline end to end and produces measured results. Its scope is deliberately limited and stated plainly: a workstation platform (no constrained hardware), the real IEEE 8500-node feeder [5] as a first benchmark run but not yet the PNNL 9500 system, a single solver (OpenDSS, no GridLAB-D cross-check), load-state snapshots rather than a full seasonal time series, one PV penetration level, a fixed rather than per-state-optimized attacker, and a Python IEEE 2030.5-style client in place of the EPRI C client. These results establish that the mechanism and the measurement work; the pre-registered full-scale sweep is not replaced by them. The cyber-side engine, credential service, OpenDSS feeder pipeline, and availability fault injection are implemented and measured; native EPRI-client traffic, constrained-hardware attestation, the GridLAB-D cross-check, the HELICS federation, interoperability tests, and the full pre-registered sweep are not.

Throughout, the primary physical endpoint is the *induced overvoltage area* ΔJ_V : the attack-case value of the one-sided overvoltage term of J_V , summed over all energized nodes as $\sum_i \max(0, V_i - 1.05)$, minus its value in the base case at the same load state. Positive values mean the attack pushed nodes above the ANSI C84.1 upper band [16]; values at or below zero mean it did not. A single power-flow solution therefore yields ΔJ_V in per-unit nodes (p.u.-node), and the time-series experiment reports the same quantity integrated over a one-minute time step, in per-unit node-minutes (p.u.-node-min). We label each table and figure with the unit it carries.

Functional invariants and containment (C1, C2)... The credential service uses real X.509 / ECDSA certificates and an EAT-style attestation token. Ten implemented security invariants all pass on the prototype: bootstrap identity cannot authorize control, issuance requires a valid fresh attestation, the certificate is bound to a freshly generated operational key, replayed evidence is rejected, an expired credential cannot issue commands, per-command scope matches issuance, a bad measurement yields only a narrowed safe-mode scope, a prior

Table 4: Security invariants implemented by the prototype (real X.509/mTLS). All ten pass. The identifiers emitted in `results/credsvc.json` are now identical to the IDs below; an earlier release numbered them I1–I5 and I7–I11, skipping I6, and the artifact has been renumbered to remove that discrepancy.

ID	Invariant	Result
I1	Bootstrap identity cannot authorize control	pass
I2	Issuance requires a fresh attestation	pass
I3	Credential bound to a fresh public key	pass
I4	Replayed evidence rejected	pass
I5	Expired credential cannot issue a command	pass
I6	Scope enforced per command	pass
I7	Bad measurement yields degraded/denied scope	pass
I8	Prior epoch cannot self-renew	pass
I9	Copied prior key fails after rotation	pass
I10	Active session/command bounded past expiry	pass

epoch cannot self-renew, and a copied prior operational key is contained at the next epoch. With per-command revalidation the post-expiry acceptance gap is $\Delta_{\text{expiry}} = -0.5\text{ s}$ (the last accepted adversarial command precedes expiry), against $+5.0\text{ s}$ without enforcement, where it grows to the cached session age. A fresh operational key is generated each epoch, so a copied key is useless after rotation. Table 4 lists the invariants, the identifier each carries in the released result file, and their outcomes. These are the invariants the prototype implements and tests; they are not a proof of the design, and properties outside this set are untested.

Overhead (C3). On the workstation tier, over 500 iterations, median software-path latencies are 0.03 ms for key generation, 0.06 ms for attestation construction, 0.23 ms for a full attested issuance, and 2.3 ms for a real mutual-TLS handshake, and the issuer sustains about 4900 renewals per second single-threaded. Renewal is therefore inexpensive here. These are standard EC P-256 primitives dominated by one ECDSA signature and verification, so constrained-controller

cost can be estimated from public-key throughput; a hardware-backed measurement is future work. These are software-path timings, not deployment latencies.

Availability (C6).. The renewal safety margin $T_{\text{TTL}} - (T_{\text{renew,p99}} + T_{\text{net}})$ stays positive at every tested lifetime down to five minutes and a 100,000-device fleet re-mints in about 20 s, so short lifetimes are latency-feasible. The risk is instead a verifier outage longer than the lifetime. Injecting verifier outages, fail-closed denial retains 0.77 of legitimate control availability, a bounded grace period 0.83, and degraded-scope safe mode 1.0. The injection is in-process; networked outages, clock skew, and renewal storms remain.

Time-series containment: what the lifetime mechanism does, and what its ratio is not (C5).. The question this experiment answers is whether bounding a credential’s lifetime bounds the physical harm a compromise does. A compromised credential drives the DER from minute five of a sixty-minute horizon on the real IEEE 8500-node feeder, and the credential’s lifetime determines whether the malicious setpoint keeps reaching it. Every (state, seed, policy) arm is an *independent* feeder compile with a fresh PV fleet and fresh regulator and capacitor state, at twenty paired seeds and two operating states. That isolation matters: the predecessor of this experiment compiled the feeder once per seed and iterated policies against the same live circuit, so each arm warm-started from the previous arm’s excursion, and the numbers reported in an earlier version of this work (6956, 7030, 3196, 3.8) inherit that leak. **We withdraw all four.** Non-convergence is handled by retrying at most twice with a fourfold control-iteration budget; 24 solves were flagged and 48 retried across the whole sweep, and unsettled solves are retained and flagged, never dropped.

Three findings, at the conformant IEEE 1547 Category B envelope of 5.28 kvar — the envelope an operator must authorize to receive volt-var at all (Table 5).

First, scope alone does not contain, and the legacy arm never recovers. A long-lived credential confined to that envelope holds the feeder in violation for the entire horizon in 20/20 seeds, with a median integral of 5656 p.u.-node-min

(bootstrap CI [5356, 5857]) under light load. Confining a compromise to the volt-var function set is not a containment measure.

Second, the lifetime mechanism works by truncation, not by attenuation, and recovery is prompt and near-complete. While the malicious setpoint is applied, the instantaneous excess overvoltage area holds at a mean of 101.8 p.u.-node (SD 10.4 across seeds); the paired per-minute accrual under the CONTAINER arm and under the legacy arm is **bit-identical over the window they share** (maximum absolute difference 0.0 across all twenty seeds and both envelopes). Lifetime enforcement changes nothing about how fast harm accrues — only when it stops. What it does change is measured and could have come out otherwise: at credential expiry the feeder returns to its pre-attack baseline within one minute in 17/20 seeds, the settled post-expiry excess has median 0.000 p.u.-node, and the entire post-expiry contribution is 1.74% of the integral. Nothing latches. A regulator or capacitor left in an excursion-driven position could have held voltage up indefinitely after the credential died, which would have made lifetime enforcement useless; it does not.

Third — and this is the point at which we part company with the earlier reading — the resulting “2.16× reduction” is not an effect size. Because the accrual rate is identical across arms and the residual is negligible, the integral is the rate multiplied by the number of minutes the credential is honoured. The ratio of the legacy integral to the CONTAINER integral is therefore pinned to the ratio of exposure windows, $(60 - 5)/25 = 2.20$, net of the recovery tail. The data behave exactly so: the observed mean ratio is 2.159 (CI [2.131, 2.183]) at the conformant envelope and 2.159 (CI [2.128, 2.184]) at the narrower 3.6 kvar envelope — *the same number to four significant figures at two different operating envelopes whose accrual rates differ by 43.7%* — and the largest ratio observed in any seed, state or envelope is 2.1988. A physical mechanism does not return the same ratio at two different physical operating points; arithmetic does. **We therefore do not report 2.16× as a measured containment factor.** The defensible statement is the relation behind it: integrated harm is the product of a measured accrual rate and the credential lifetime, up to the exposure window,

so the containment factor an operator obtains is $(\text{exposure window})/T_{\text{cred}}$ — a quantity the operator chooses, not one we measured. Quoting the 2.16 would be quoting our own choice of a twenty-five-minute lifetime against a sixty-minute horizon back to the reader. The lifetime sweep below measures the relation itself.

The honest counter-arm. At the non-conformant 0.6 kvar envelope the ordering reverses: the CONTAINER arm is *worse* than the legacy arm (median integral 20.1 against 12.4; ratio 0.622, CI [0.455, 0.781]; 0/20 seeds favour CONTAINER, and under normal load 0/20 at ratio 0.083). Where the authorized envelope is already physically inert, the credential lifecycle buys nothing and the transition at expiry costs something: regulator tap operations rise from a median of 78.5 to 125, a 59.2% increase. The same cost appears at the conformant envelope, where taps rise from 67 to 101.5, a 51.5% increase in regulator wear that the voltage endpoint does not price. We report this because it is the boundary of the mechanism’s usefulness: lifetime enforcement pays only where scope cannot contain, which — per Table 5 — is precisely the conformant case, but it is not free.

Two arms we withdraw as uninformative. The mechanism ablations A2 (session enforcement removed) and A3 (command cleanup removed) are **not identified by this harness**. Its activity predicate maps the `legacy`, `no_session` and `no_cleanup` lifecycles onto the same branch, so A2, A3 and B1 are one arm under three labels: their twenty-seed output series are byte-identical, in both operating states. Likewise A4 and B3 share a scope and a lifecycle and are a single arm. The earlier claim that removing session enforcement or command cleanup “returns the feeder to no recovery, so each mechanism is necessary” was therefore reading a relabelling as a result, and we withdraw it. The per-command enforcement these ablations were meant to probe *is* tested, but on the cyber side, by invariants I5, I6 and I10 and by the $\Delta_{\text{expiry}} = -0.5\text{s}$ measurement above; the feeder harness cannot distinguish them and we no longer ask it to.

A label correction. The arm this artifact calls “full scope” commands (12 kW,

3.6 kvar) while the arm it calls “narrow ACL at IEEE 1547” commands (12 kW, 5.28 kvar), and the inverter’s own reactive limit is 5.28 kvar. The “narrow” envelope is thus a strict *superset* of the “full” one, which is why the legacy arm reads 5656 at the conformant envelope against 3851 at the narrower one. That is not narrowing making things worse; it is an inverted label, and no comparison in this paper reads a scope *ordering* off those two arms. Table 5, which sweeps the envelope continuously and names each level by its commanded kvar, is the evidence for scope.

Feeder consequence across operating states (C5)... We run an export/overvoltage attack on the real IEEE 8500-node feeder (4876 buses, 8531 nodes, 1177 loads, 10.8 MW aggregate load at nominal multiplier) with 600 seeded rooftop-PV units of 12 kW each (7.2 MW), ten paired seeds, under three load states. Because the 8500 base has native undervoltage at its extremities, we report the induced *overvoltage* area the attack targets rather than a two-sided area that would credit the attack for lifting sagging nodes; this attack-specific endpoint was chosen after observing the base undervoltage, while the two-sided area remains the declared primary system-level endpoint. Throughout, ΔJ_V is measured against *legitimate scheduled PV operation under an IEEE 1547-2018 Category B volt-var curve*, not against a feeder with no DER.

How narrow must scope be before it stops containing? (C5). This is the only question an operator actually needs answered, and fixing two envelopes by fiat cannot answer it. We therefore sweep the authorized envelope continuously along the two axes an IEEE 2030.5 **FunctionSetAssignments** narrowing can act on, at five paired seeds and two operating states, and locate the breach threshold (Table 5).

On the *active-power* axis, where the adversary holds **DERControl**, there is a genuine threshold. Under light load, ΔJ_V stays negative up to $\sigma=0.7$ of nameplate and turns positive between $\sigma=0.7$ and $\sigma=0.8$; the IEEE 1547 overvoltage trip screen fires in 60% of seeds at $\sigma=0.7$ and in every seed from $\sigma=0.8$. Restricting active-power authority below roughly 70% of nameplate does contain.

On the *reactive* axis it does not, and this is the paper’s most consequential result. Confining the adversary to the volt-var function set — the narrowest scope that still delivers the service IEEE 1547-2018 requires and CSIP volt-var programs procure — **breaches at every envelope we tested, including an envelope of zero.** At the full Category B capability of 5.28 kvar (44% of nameplate) the median induced area is 87.9 (per-seed range 79.9–100.8, all five seeds tripping OV1 at ≥ 1.10 p.u., $v_{\max}$ up to 1.142). At 0 kvar it is still 31.2, with every seed tripping. The mechanism at zero is worth stating plainly: an adversary authorized only to configure volt-var need not inject anything: it is enough to *suppress* the reactive absorption the feeder was relying on, which is squarely inside a volt-var-only authorization. **There is therefore no reactive envelope that both contains a compromise and delivers the volt-var service.**

This overturns the reading in an earlier version of this work. That version placed “narrow” at 0.6 kvar, which is 5% of a 12 kW unit’s nameplate and only 11.4% of the Category B range, and reported that it held the feeder inside the ANSI band. It does not deliver volt-var at that envelope, so the containment it appeared to show was the containment of authorizing almost nothing. Measured against legitimate operation rather than against an empty feeder, even that envelope breaches ($\Delta J_V = 17.4$, trips in 4/5 seeds). We accordingly withdraw the claim that narrow scope “retains the most common legitimate DER function while bounding the attack”: at the tested envelope it did not retain the function, and at the conformant envelope it does not bound the attack.

Consequence is strongly state dependent. Under normal load the same sweep produces a median $\Delta J_V \leq 12.0$ on the active axis and ≤ 11.4 on the reactive axis, with no trips at any envelope in any seed. Light load with high PV is the binding state, as the physical-state dependence of Section 3 anticipates.

Why ΔJ_V is negative at small envelopes.. The negative entries on the active axis are not noise and not reactive “support”. Against a legitimate counterfactual in which PV generates at its full scheduled output, an adversary restricted to

$\sigma < 0.7$ of nameplate is *curtailing* relative to that schedule: it injects less active power than legitimate operation would, so the voltage rise along the feeder falls and the one-sided overvoltage area shrinks. A restrictive active-power scope therefore turns the compromise into an unwanted-curtailment problem — an availability and revenue harm our voltage endpoint does not price — rather than an overvoltage problem. An earlier version of this work attributed such negative values to a narrowed reactive envelope “marginally supporting” voltage, which is backwards: voltage support raises voltage and would *increase* a one-sided overvoltage area. The correct mechanisms are curtailment relative to schedule on the active axis and reactive *absorption* on the reactive axis.

This attack is loud, and that matters.. At the full-scope light-load condition, a median 6662 of 8531 energized nodes (78%; per-seed range 6184–7104 over ten seeds) sit above 1.05 p.u., with a median maximum voltage of 1.124 p.u. rising to 1.135 p.u. That is not a stealthy manipulation; it is a feeder-wide excursion that AMI voltage reporting, regulator operation counts, and customer complaints would surface within a single metering interval. Two conclusions follow. First, it weakens any narrative in which such a compromise quietly persists for the two-year lifetime of a certificate: an adversary behaving this way is found quickly, and the operative containment latency becomes the utility’s detection-and-response time rather than the credential lifetime. Second, and more usefully, it identifies the attacker our evaluation does *not* cover. An adversary who holds voltages just inside the ANSI band, or who acts only during the minutes of peak PV, would extract far more value from long persistence precisely because nothing alarms — and that is exactly the case in which credential lifetime, rather than detection, is the binding constraint. Our pre-registered protocol specifies an operationally realistic telemetry-limited attacker alongside the oracle worst case; the stealth-constrained variant is the more important of the two for this paper’s thesis, and it is not yet run. We regard this as the most significant gap in the present evaluation.

Table 5: Scope-envelope sweep on the IEEE 8500-node feeder: median induced overvoltage area ΔJ_V (p.u.-node) against legitimate volt-var operation, and the fraction of seeds in which the IEEE 1547 Category II overvoltage trip screen fires. Five paired seeds per cell. The reactive axis holds active power at its legitimate schedule and varies only the authorized reactive envelope.

Authorized envelope	light load / high PV		normal	
	ΔJ_V	trip	ΔJ_V	trip
<i>Active-power axis</i> ($\sigma \times$ nameplate, DERControl)				
$\sigma = 0.0$	-56.0	0.00	-0.07	0.00
$\sigma = 0.5$	-30.3	0.00	-0.33	0.00
$\sigma = 0.7$	-1.3	0.60	0.17	0.00
$\sigma = 0.8$	22.1	1.00	1.49	0.00
$\sigma = 1.0$	63.1	1.00	11.98	0.00
<i>Reactive axis</i> (volt-var scope; legitimate P retained)				
0 kvar (suppress volt-var)	31.2	1.00	2.01	0.00
0.6 kvar (earlier “narrow”, 11% of 1547)	17.4	0.80	1.38	0.00
2.4 kvar (45% of 1547)	49.9	1.00	5.58	0.00
5.28 kvar (full IEEE 1547 Cat B)	87.9	1.00	11.44	0.00

Results measured under the superseded scope configuration.. Three further sweeps were run before the scope-envelope result above, using the earlier 0.6 kvar “narrow” envelope and the zero-DER counterfactual. We report them, and we mark plainly what they do and do not now support. Under a legacy full-authorization compromise, induced overvoltage grows with PV penetration from 8 at 20% to 254 at 100% (five seeds, Fig. 3); it is reachable through active or reactive authority alike, with areas of 122 (export-limit), 97 (volt-var), and 57 (volt-watt) across attack families (five seeds, Fig. 4); and it appears on a second public feeder, the IEEE 123-bus system, at up to 2.2 under high penetration (three seeds), though that stiffer regulated 4.16 kV feeder needs more penetration to breach the band. **These legacy-baseline measurements stand.** What does *not* stand is the accompanying claim, made in an earlier version, that the

narrowed scope held induced overvoltage at or near zero in every one of these cells, and that it therefore “retains the most common legitimate DER function while bounding the attack”. That envelope is 11.4% of the IEEE 1547 Category B reactive range and does not deliver volt-var, and the near-zero readings were taken against a counterfactual feeder carrying no DER. Table 5 supersedes them. We have not re-run the penetration, attack-family, or IEEE 123 sweeps at the conformant envelope, so the conformant-scope result is established at one penetration level, on one feeder, in the two operating states of Table 5.

Modeled retained authority, and what it is not evidence for (H2).. An earlier version of this work summarised the baselines with a proxy $E_{\text{phys}} = \Delta J_V \times \text{BR}_{\text{auth}}$ and reported that moving from ephemeral to legacy persistence “multiplies light-load exposure roughly $1700\times$ ”. **We withdraw that figure.** The two cells share the same authorization scope and therefore the same measured ΔJ_V of 126.60, so that factor cancels exactly and the ratio reduces to $17524/10 = 1752.4$, a quotient of two assumed lifetimes. It contains no measurement whatsoever, and presenting it as an evaluation result was wrong. The only defensible statement is the tautology that a credential assumed to last two years is assumed to be held about 1750 times longer than one assumed to last ten hours. The direct, measured replacement for that claim is the time-series contrast above.

We also rename the quantity. It is E_{phys} , a physical-consequence-times-duration product in p.u.-node-hours, and it is *not* the flexibility-time BR_{exp} of Section 3, which carries kW·h and kVAr·h. The two were previously distinguished in prose while sharing a name. This paragraph belongs to hypothesis H2 (capacity-time exposure); an earlier version labelled it H1, which the pre-registration reserves for the lifetime-by-scope interaction on voltage-violation area.

Two entries in Table 6 contradict the narrative that earlier version drew from it, and we now say so directly.

First, **B5’s retained authority (12.22 h) is longer than B3’s (10.0 h).**

On that dimension the full design is strictly worse than ephemeral-only. The cause is not an error: B3’s credential simply expires and is never renewed, whereas B5’s is attestation-gated and *does* renew, surviving a $\text{Geometric}(p)$ number of cycles until attestation catches the compromise. With $T=6$ h and $p=0.5$ the expectation is $T/p = 12$ h, which is what the model returns (and at $p=1$ it returns exactly 6 h). Renewal is what makes CONTAINER deployable — an un-renewed credential strands the legitimate device too — and its price is that containment latency is governed by detection probability, not by lifetime. This is the T/p figure of merit of Section 4, and it means B5’s advantage over B3 comes from scope, not from time.

Second, **B2 and B5 both sit at ≈ 0 in this table, and we no longer read that as support for the design.** It is the arithmetic consequence of a “narrow” scope whose measured ΔJ_V was 0.0 exactly: any lifetime multiplied by zero is zero. Once scope is set at the conformant envelope of Table 5, the physical factor is not zero and the tie disappears. **We therefore withdraw the sentence “neither half alone suffices” as it was argued from this table**, since the comparison it invoked — B5 against B3 and B4 — excluded the one baseline that tied B5. The claim is re-established, on different and measured grounds, by the time-series contrast at conformant scope reported above: there, scope alone does not contain, which is what makes duration enforcement necessary.

Seed inventory. Table 7 is the complete inventory of every measured run behind this section, so that no experiment’s scale has to be inferred from prose.

Limitations. These results are pilot results and we do not generalize them. Physical results come from two public feeders and a single OpenDSS solver, with no GridLAB-D cross-check and a fixed export attacker; Table 7 gives the full seed inventory.

The IEEE 9500 failure mode, stated. The PNNL 9500-node system was obtained but did not converge in our build, and we name the mode rather than leaving it an unexplained absence. The failure is in the *base case*, with no

Table 6: Modeled retained authority BR_{auth} and the exposure product $E_{\text{phys}} = \Delta J_V \times \text{BR}_{\text{auth}}$ in p.u.-node-hours (light load, IEEE 8500, $n=10$ seeds). ΔJ_V is measured; BR_{auth} is a *model output* under assumed lifetime and detection parameters, not a measurement, so every product in the last column inherits that assumption. The ≈ 0 entries reflect a non-conformant narrow envelope whose measured ΔJ_V was zero; see Table 5.

Baseline	Scope	BR_{auth} (modeled)	E_{phys}
B1 legacy full	full	$\sim 2 \text{ yr}$	2.2×10^6
B2 ACL-narrow	narrow (non-conformant)	$\sim 2 \text{ yr}$	≈ 0
B3 ephemeral-only	full	10.0 h	1.3×10^3
B4 attestation-only	reference	$\sim 14 \text{ d}$	8.3×10^3
B5 full CONTAINER	narrow (non-conformant)	12.22 h	≈ 0
B6 safe-mode	narrow (non-conformant)	12.22 h	≈ 0

attack applied and no PV of ours placed: running the model’s own shipped driver unaltered — which sets the voltage bases, calls `calcv`, and solves at `maxiter=1000 maxcontroliter=100` — returns `Converged = False` with an empty error description, and every bus per-unit magnitude reads zero, i.e. the solution collapses rather than merely missing tolerance. It reproduces at 1000 iterations, under both the normal and Newton algorithms, and with feeder controls both enabled (`controlmode=static`) and disabled (`controlmode=off`), so it is neither a control-loop oscillation nor an iteration-budget shortfall. Our build is DSS C-API 0.14.5 / DSS-Python 0.15.7 / OpenDSSDirect.py 0.9.4 on macOS arm64. **We did not resolve it and we do not attribute it:** we cannot exclude a platform or build defect on our side rather than a defect in the model, and we did not have a second OpenDSS build available to discriminate. It is deferred to an appendix tier per the pre-registered contingency. The consequence for this paper is that the cross-feeder generality of these results rests on the IEEE 123 confirmatory run alone, which is weak, and we do not claim more. Retained-authority figures, and the exposure proxy built on them, are model outputs under assumed attestation and lifetime parameters, not measured field values. Overhead is workstation-only: there is no constrained-

Table 7: Complete inventory of measured runs. Seeds are paired across policy arms within an experiment. “Arms” counts the policy or envelope conditions evaluated at each seed.

Experiment	Seeds	Arms	States
IEEE 8500 snapshot sweep	10	2	3
IEEE 8500 time series (corrected)	20	9	2
Scope-envelope sweep	5	21	2
PV penetration sweep	5	1	1
Attack-family sweep	5	4	1
Legitimate-utility run	3	3	1
IEEE 123 confirmatory run	3	2	1
Credential-service invariants (C1)	1	10	—
Overhead microbenchmarks (C3)	500 iterations	4	—
Availability fault injection (C6)	3 policies	3	—

hardware or TEE measurement. The availability result injects verifier outages in-process rather than as networked or hardware failures, and clock skew and renewal storms are untested. Incremental deployment is by construction; interoperability against an unmodified IEEE 2030.5 endpoint is untested, and the exact clause-level delta between the 2018 and 2023 editions of the standard’s certificate-management text remains open pending access to the revised standard. CONTAINERDER bounds physical consequence under the evaluated scope configurations; the pre-registered sweep on two public feeders, with a cross-solver check, seasonal states, constrained hardware, and the full seed budget, is what would support quantitative claims beyond these.

9. Related Work

We position CONTAINERDER against ten strands (Table 8).

IEEE 2030.5 and CSIP security. The 2018 and 2023 editions of IEEE 2030.5 mandate mutual TLS with X.509 device certificates [7, 8], constrained for interconnection by the Common Smart Inverter Profile [9]. The closest prior work is the 2030.5 security tutorial of Passos et al. [6], a survey that proposes no

redesign, quantifies no feeder consequence, and formalizes no impact metric; Qi et al. survey the broader DER and smart-inverter surface [18]. The Sandia analysis supplies the revocation constraints we respect [1].

DER and smart-inverter attacks. High-wattage IoT botnets can move the grid [19, 20]; on the DER side, inverter attacks destabilize grid and market [21] and distribution voltage control is attackable under high PV [22], while the SUN:DOWN disclosure catalogs fleet-scale inverter vulnerabilities [23]. We tie the attack surface to the credential lifecycle rather than assuming device compromise.

Distribution-grid cyber-physical impact. False-data-injection work quantifies compromise consequence [24, 25, 26], grounded by the 2015 Ukraine incident [27] and quantitative risk formulations [28]; our feeder method is supporting, not novel.

Short-lived certificates and PKI lifecycle. ACME [29] and Let’s Encrypt [30] automate issuance, SCEP [31] is the incumbent device enrollment, and HTTPS-ecosystem measurement quantifies the lifecycle problem [32]; we adopt short lifetimes as one link of a chain.

Hardware-bound identity. Secure device identity [12], firmware TPM [33], and hardware DICE [34] root our bootstrap layer.

Remote attestation. We use the RATS architecture [14], the Entity Attestation Token [13], and BRSKI [35], on established principles [36], adding fresh-key binding.

Attack-graph and cyber-physical metrics. Logic-based [2, 37, 38], probabilistic and Bayesian [3, 11, 10], and k -zero-day [4] metrics are static-snapshot and cyber-only; Section IV shows what reachability-only reading omits, positioning our model as complementary.

Co-simulation. HELICS federates the cyber and grid models [39, 40] with GridLAB-D on the power side [41].

DER authorization. IEEE 1547 [17] defines the DER functions, DERMS perform feeder voltage management [42], and role-based access control [43] abstracts the scoping we narrow and enforce.

Revocation at scale. Web-PKI revocation delivery is gap-ridden [44], scalable revocation remains an open problem [45, 46], and even OSCP Must-Staple adoption is weak [47], which motivates containment by short-lived credentials rather than by revocation.

Workload identity: the closest existing system. SPIFFE and its SPIRE runtime [48, 49] implement almost exactly the chain we describe. SPIRE attests node and workload, issues short-lived scope-bound X.509 SVIDs keyed to the attested identity, refuses renewal when attestation fails, and relies on expiry rather than revocation; recent work extends it to IoT fleets [50] and nested tokens [51]. We must therefore answer whether CONTAINERD is workload identity with a DER label. Two differences are load-bearing. First, SPIFFE has no notion of a *physical effect that outlives the credential*: revoking a workload identity stops a process opening new connections, but does not retract a volt-var curve already written into an inverter or a schedule already accepted for tomorrow. Our T_{sess} and T_{cmd} enforcement exists because actuation persists in a way computation does not. Second, scope binds to **FunctionSetAssignments**, whose conformant settings are dictated by IEEE 1547-2018 [17] and CSIP [9] rather than chosen freely — and our central negative result is that this externally fixed envelope is too wide to contain, which has no analogue where the deployer picks the scope. The field constraints of Section 4 differ too.

The Web PKI has already adopted non-renewal. Our mechanism is not speculative, and we should not present it as though it were. The CA/Browser Forum defined a Short-lived Subscriber Certificate in 2023 and *exempted it from CRL and OSCP status checking* [52], which is precisely “non-renewal replaces revocation”, already ratified; Baseline Requirements v2.2.8 sets that threshold at seven days [53]. Ballot SC-081v3, approved in April 2025, phases the maximum TLS certificate validity from 398 days down to 47 days by 2029 [54]. This is simultaneously the strongest external validation of our premise — the largest deployed PKI reached the same conclusion independently — and the sharpest limit on our novelty. We claim no novelty for short-lived credentials or for expiry-based containment. What we claim is the mapping onto DER authoriza-

Table 8: Positioning against representative prior work. y = yes, n = no, p = partial, – = n/a.

Work	2030.5	Scope	Time	Phys.	Attest.	Proto.	Feeder
Attack-graph metrics [2, 3, 4]	n	p	n	n	n	–	n
2030.5 security survey [6]	y	p	n	n	n	n	n
Attestation building blocks [14, 13, 12]	n	p	p	p	y	p	n
CONTAINDER (this work)	y	y	y	y	y	p*	p*

y = addressed; p = partial; n = not addressed; – = not applicable. *Partial: a prototype and a pilot feeder evaluation are reported in Section 8; native-client interoperability, constrained hardware, and the full pre-registered sweep are not yet done.

tion: which scope object to bind, what the conformant scope envelope costs, and that session and command-effect enforcement are separately necessary because grid actuation persists.

DER guidance and competing key management. IEEE 1547.3-2023 is the cybersecurity guide for interconnected DER [55] and the natural home for these recommendations; NISTIR 7628 [56] and IEC 62443 [57, 58] give the enclosing frameworks; NREL’s DER certification line [59, 60] and gap analysis [61] document the certification landscape. IEC 62351 is the sharpest comparison: Part 8 gives role-based access control [62] and Part 9 specifies key management *including* enrollment and revocation for power-system equipment [63]. Part 9 is a directly competing answer, and the honest question is why 2030.5/CSIP does not adopt it. Not technical superiority on our side: 62351-9’s revocation machinery assumes the connectivity and responsible-reporting party the Sandia analysis [1] found absent, and CSIP deployments certify against the 2030.5 profile. Where an operator has that connectivity, 62351-9 is a reasonable alternative, and we say so rather than claim the field.

10. Ethics and Responsible Disclosure

This work studies the consequences of credential compromise in order to bound them, and it carries a corresponding responsibility. All experiments use synthetic and lab-generated certificates. We use no real vendor keys or secrets, we test against no production DER fleet or any internet-reachable device, and we build no exploit tooling bound to a specific vendor product. The malicious dispatch programs exist only inside the feeder co-simulation and are never emitted to hardware.

The physical-consequence results describe what a compromised but protocol-conformant credential can already do under the deployed standard; they do not introduce a new vulnerability so much as measure an existing exposure that the standard’s prohibition on revocation leaves unmitigated. The redesign and its migration path are offered as the constructive counterpart to the measurement: the point of quantifying the blast radius is to shrink it.

Disclosure and release, reconciled.. These two commitments are one timeline, not two. Nothing in this work identifies a vulnerability in a specific vendor product; what it identifies is a property of a published standard, which the standards body and the Sandia analysis [1] already describe in public. There is accordingly no embargoed finding whose disclosure must precede release. The artifact — code, result files, and pre-registration — is deposited and citable at the time of publication, which is the same event as public release. Should the confirmatory sweep identify a concrete weakness in a specific implementation, we will notify that implementer and the relevant standards body first and delay only the affected component of the release; the remainder is not held back.

11. Limitations and Conclusion

Limitations.. The boundaries of the evaluation are stated next to the results they bound, at the end of Section 8: two public feeders and a single solver, one PV penetration level at the conformant scope envelope, modeled rather than

measured retained authority, workstation-only overhead, in-process availability injection, untested interoperability, an unmeasured stealth-constrained attacker, and no access to the 2023 edition of the standard. Two further boundaries belong to the model. The authorization graphs used by the engine are synthetic, because real utility authorization graphs are not public; we mitigate this with a released parameterized generator and by ablating access-model permissiveness rather than fixing it. And our claims about revocation rest on the Sandia analysis rather than a clause-level reading of the edition in force.

Conclusion.. Credential compromise in IEEE 2030.5 is not well described by how many nodes an adversary reaches. It is described by how much physical capacity those nodes command, how long the adversary keeps it, and what the feeder does in response. Our central empirical finding is a negative one, and it constrains the design space more than a favourable result would have. **On the IEEE 8500-node feeder there is no authorization scope that both contains a credential compromise and delivers the volt-var service IEEE 1547 requires.** An adversary confined to the conformant reactive envelope still drives the feeder past an overvoltage trip threshold in every seed, and an adversary that merely suppresses volt-var — needing no injection authority at all — does so too. Least privilege, applied to this function set on this feeder, is not sufficient.

What follows is not that the mechanism is useless but that its useful part is temporal. Once scope cannot bound the instantaneous envelope, bounding the *duration* of retained authority is what remains, and that is what attested short-lived credentials with session and command-effect enforcement provide. The price is stated rather than hidden: containment latency is the credential lifetime divided by the per-cycle attestation detection probability, not the lifetime itself, and a gateway-proxy tier is the only one of our three deployment tiers that resolves the secure-time and constrained-hardware constraints the mechanism inherits from the problem it addresses. These are pilot results on one feeder; the pre-registered confirmatory sweep is what would generalise them.

Data availability

The analysis engine, the X.509/mutual-TLS credential service, the OpenDSS feeder experiment scripts, the pre-registration document, and every result file underlying the numbers reported here are provided as a single archive accompanying this submission, and will be deposited under a persistent identifier on acceptance; the archive is cited as reference [64]. Each reported numeral is checked against that archive by `check_numbers.py`, which fails if a numeral in the manuscript body cannot be located in the released result files. The IEEE 8500-node and IEEE 123-bus test feeders are publicly available from the referenced OpenDSS distributions. The one input we cannot redistribute is the IEEE 2030.5-2023 standard text itself, which is sold by IEEE; Section 2 states precisely which claims depend on it and which edition each was checked against.

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Declaration of competing interest

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CRediT authorship contribution statement

Huan Bui: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used a large language model to assist with copy editing, LaTeX table formatting, and the drafting of code comments in the released artifact. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. No generative AI tool was used to produce, analyse, or interpret any experimental result, and no text generated by such a tool was incorporated without verification against the underlying data.

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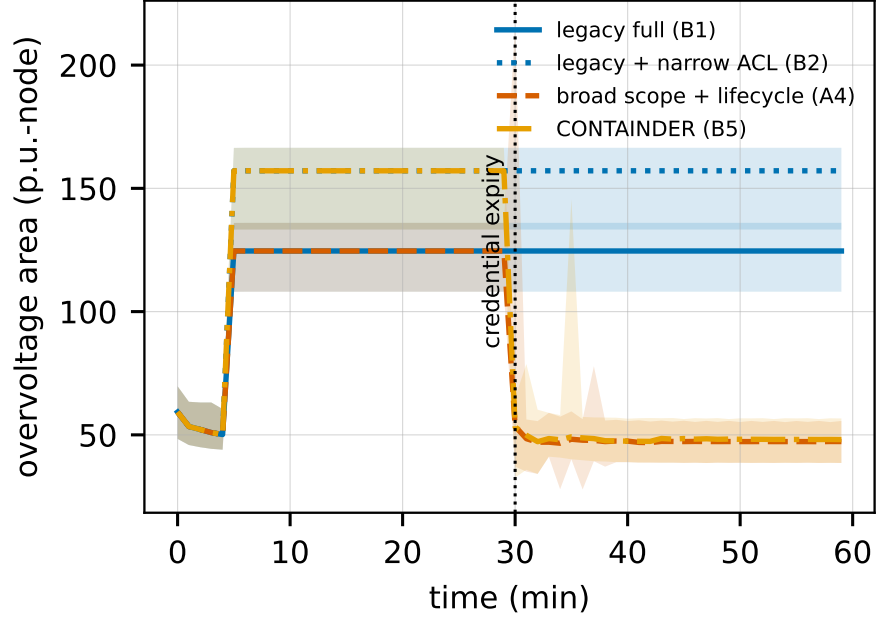


Figure 2: Overvoltage area (p.u.-node) against time on the IEEE 8500-node feeder, light load: median across $n=20$ paired seeds, with the min-max band over those seeds shaded. Each arm is an independent feeder compile. The two legacy arms never recover within the horizon (20/20 seeds each). The two lifetime-bounded arms track their legacy counterparts *exactly* until the credential expires at minute 30 and then return to baseline within one minute — the mechanism truncates the exposure window rather than attenuating the excursion, so the ratio between the integrals is set by the lifetime chosen relative to the horizon and is not an effect size (see text). B2/B5 sit above B1/A4 because the envelope labelled “narrow ACL” is 5.28 kvar and the one labelled “full” is 3.6 kvar; the labels are inverted in the artifact.

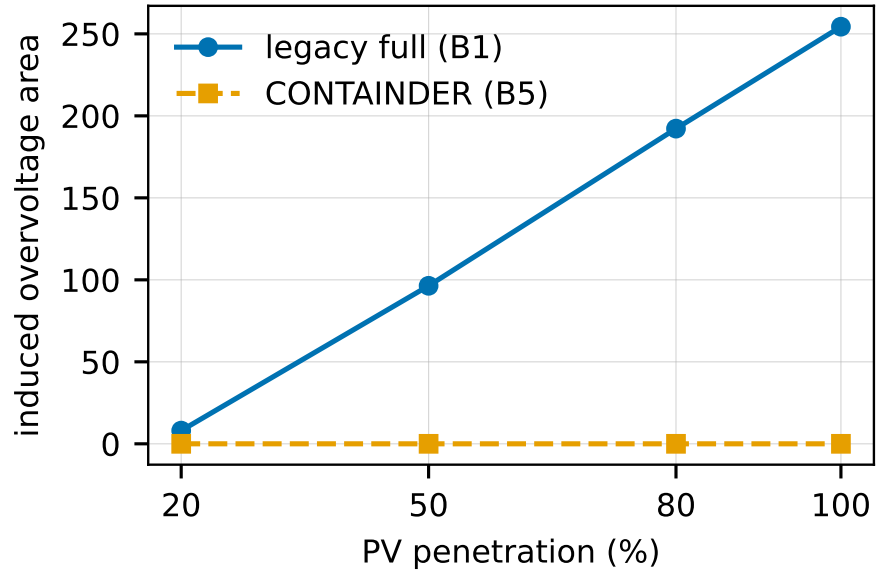


Figure 3: Induced overvoltage area (p.u.-node, snapshot) versus PV penetration (IEEE 8500, light load, $n=5$ seeds). **Only the legacy trace stands.** Harm scales with penetration under the legacy baseline. The flat CONTAINER trace was measured at the superseded 0.6 kvar envelope, which Table 5 shows is physically inert and does not deliver volt-var, so it reads zero at every penetration by construction and is *not* evidence of containment. We have not re-run this sweep at the conformant envelope.

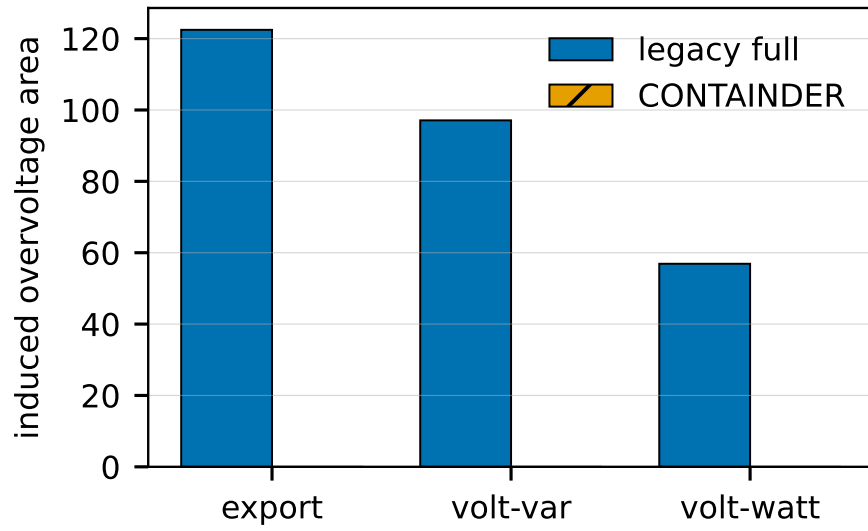


Figure 4: Induced overvoltage area (p.u.-node, snapshot) by attack family (IEEE 8500, light load, $n=5$ seeds). **Only the legacy bars stand:** overvoltage is reachable through active or reactive authority alike. The CONTAINERD bars were measured at the superseded 0.6 kvar envelope and read zero because that envelope is inert (Table 5), not because narrowing contained the attack.